12–17 第二篇

Weather Forecasting: Science Behind the Forecast

We check the weather every day—before a trip, a picnic, or just to decide what to wear. But how do forecasters know what the weather will be? Weather forecasting combines science, technology, and experience. Satellites, radars, and weather stations collect huge amounts of data from the atmosphere. Computers run models that simulate how the atmosphere will behave. Forecasters use these tools to make predictions. [1]

Weather systems are complex. High and low pressure areas, fronts, and wind patterns all affect the weather. Warm air rises; cold air sinks. Ocean currents and mountain ranges change temperature and rainfall. Small changes in one place can lead to big changes elsewhere. This is why forecasts become less accurate further into the future. Short-term forecasts of one to three days are usually reliable. [2]

Technology has improved forecasts greatly. In the past, forecasters relied mainly on basic instruments and experience. Today, supercomputers process data from around the world. Radar can detect rain, snow, and storms in real time. Satellites track hurricanes and monitor global patterns. Smartphone apps bring forecasts to our fingertips. We have more information than ever before. [3]

Forecasts help us plan our lives. Farmers use them to decide when to plant or harvest. Pilots need weather information to fly safely. Emergency services prepare for storms and floods. Even a simple "will it rain tomorrow?" helps millions of people each day. [4]

Weather forecasting will keep improving as technology advances. Better models and more data will make predictions more accurate. Yet the atmosphere will always hold some surprises. [5]

选项

A. Weather forecasting relies on data, technology, and scientific models to predict what the atmosphere will do. B. Because the atmosphere is complex and sensitive, forecasts are most accurate for the near future. C. Modern technology has transformed how we collect data and make weather predictions. D. From agriculture to travel to safety, weather forecasts serve many important purposes. E. As science and technology improve, we can expect better forecasts, though some uncertainty will always remain. F. For example, the jet stream is a fast-moving air current that influences weather patterns across continents.

答案键（Answer Key）

[1] A [2] B [3] C [4] D [5] E

深度解析

- **[1] A**：
  - **线索**：空前介绍人们每天查天气、预报如何得出、预报结合科学技术与经验、卫星雷达气象站收集数据、计算机运行大气模拟、预报员据此预测，空后需总结预报的基础。A的"relies on data, technology, and scientific models to predict what the atmosphere will do"与前文对应。
- **[2] B**：
  - **线索**：空前说天气系统复杂、高低压锋面风场影响、冷暖空气运动、洋流山脉改变温雨、小变化可引发大变化、因此远期预报准确性下降、短期一至三天通常可靠，空后需总结复杂性与预报时效。B的"Because the atmosphere is complex and sensitive, forecasts are most accurate for the near future"与前文呼应。
- **[3] C**：
  - **线索**：空前说技术大幅改进预报、过去依赖简单仪器与经验、今有超级计算机处理全球数据、雷达实时探测、卫星追踪飓风、手机应用普及、信息前所未有，空后需总结技术变革。C的"Modern technology has transformed how we collect data and make weather predictions"与前文对应。
- **[4] D**：
  - **线索**：空前说预报帮助规划、农民决定种植收获、飞行员需天气信息、应急部门应对风暴洪涝、简单问题如明日是否下雨惠及数百万人，空后需总结多领域应用。D的"From agriculture to travel to safety, weather forecasts serve many important purposes"与前文呼应。
- **[5] E**：
  - **线索**：空前说预报将随技术继续改进、更好模型更多数据提高准确性、但大气总会带来意外，空后需总结前景与局限。E的"As science and technology improve, we can expect better forecasts, though some uncertainty will always remain"与前文对应。
- **干扰项分析**：
  - **F. For example, the jet stream...**：以For example开头，为具体大气现象案例（急流）。各空均为概括或过渡句；[2]易被误选因涉及大气运动，但该空需总结大气复杂性与预报时效关系，而非补充某一具体现象细节。